

A Comparison of Scheduling Algorithms for Packet Voice in a 5G Cellular System and their Impact on Full Buffer eMBB Flows

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Abstract—Since the radio channel between the base-station and a User Equipment (UE) is stochastic, semi-persistent scheduling (SPS), commonly used for Voice over New Radio (VoNR), may be suboptimal in the utilisation of time-frequency resources, thus providing lower residual bit-rates for enhanced mobile broadband (eMBB) flows. We consider optimal scheduling of voice packets so that their resource utilisation is minimised, without violating their transmission deadlines. Assuming continuous time-frequency resources, we compare the performance of the following algorithms: SoA: Schedule on Arrival, which is one version of semi-persistent scheduling; RDR: Rate to (Residual) Deadline ratio based policy, a natural heuristic; OST: Optimal Stopping Time scheduling, which uses the finite horizon backward dynamic programming approach. We compare the above policies with GAS: Genie-Aided Scheduling, a bounding approach, in which the scheduler, non-causally, gets the resource requirements in every slot up to the deadline of the packet. We empirically evaluate the impact of these scheduling policies on VoNR resource utilisation and packet loss, and on the eMBB throughput region. We also study the effect of advancing the deadline for packet delivery, to reduce the downlink scheduling overhead and UE power consumption. For a 100 MHz system, i.i.d. channel over slots, independent across users, 100 VoNR calls, and call-by-call scheduling, we find that, for a packet deadline of 20 ms (40 slots), state dependent policies can provide up to 80% reduction in resource utilisation by packet-voice, whereas a deadline of 1.5 ms (3 slots) already yields a reduction of up to 60%. When there are slot-to-slot correlations in the channel, there is reduction in the above gains for coherence times 5 ms (10 slots) and 10 ms (20 slots), but there is still room for optimisation beyond SPS.

Index Terms—joint scheduling real-time and eMBB, optimal scheduling

I. INTRODUCTION

In this paper, we consider the joint scheduling of downlink Internet data transfers, such as file transfers or video downloads (as distinct from video streaming), and guaranteed performance packetised voice calls (as distinct from packet-voice over the Internet). The data transfer we consider is assumed to be elastic; the performance objective is the average throughput during a large file transfer; as opposed to short-term throughput requirements, as would be the case with streaming video. By “guaranteed performance” packet-voice we mean that the radio access network needs to provide certain guarantees to the voice packets, without which the end-to-end performance of the voice would be unsatisfactory.

We are motivated primarily by the eMBB and VoNR services in 5G public cellular systems. While eMBB refers to a broad range of mobile Internet services, here we consider only large volume data transfers; intermittent “web browsing” would be

another example of the use of the eMBB service, but is outside the scope of this paper. With this restriction, we model the downlink buffers of eMBB UEs to have enough data to fill an entire Transport Block (TB) with only one UE’s data whenever that UE is scheduled, i.e., the “full-buffer” model.

We consider $N+2$ UEs, out of which N are on VoNR phone calls, and the other two are doing large downlink file transfers (to be called eMBB UEs). The figure of 2 eMBB UEs is only for the purpose of assessing the impact of various policies for scheduling the VoNR calls, on the throughput capacity available to the eMBB UEs. In future work, we will consider joint scheduling of VoNR and eMBB UEs, where the number of each type will be parameters.

Each VoNR call is modeled as bringing a packet every 20 ms during an active period, followed by an inactive period. We assume that the UEs are only in the listening mode, i.e., the VoNR calls are always in the mode of sending packets downlink, alternating between activity and inactivity. This will provide an upper bound on the utilisation of downlink time-frequency resources for VoNR; if the talk-listen ratio is 50% then the obtained resource utilisation should be halved.

We consider call-by-call scheduling; joint scheduling of all the N voice calls is a topic of our ongoing research. Each VoNR packet is required to be delivered to the UE within 20 ms, i.e., 40 slots. Thus, semi-persistent scheduling (SPS), the commonly adopted technique, can schedule the first packet of an activity period in any of the 40 slots, and continue to schedule the successive packets every 40 slots thereafter. We assume that the time-frequency resource requirements of each VoNR packet in successive slots are independent and identically distributed (i.i.d.) random variables, or form a simple Markov chain which models a correlated channel. Evidently, by scheduling the VoNR packets periodically, we lose the opportunity to minimise the time-frequency resources required to transmit the packets. We study a non-causal (thus, impractical) bounding policy, called genie-aided scheduling (GAS), where a “genie” provides the scheduler with time-frequency resources required to send a packet over any of the 40 slots up to the deadline after its arrival. By modeling VoNR scheduling with a deadline as an optimum stopping time problem, we propose OST scheduling; the solution is the same as for the well-known “material purchase with a deadline” problem [2]. Since OST requires the computation of channel statistics dependent thresholds, we also study a simple heuristic that uses a tunable thresholding of a resource to remaining-

deadline ratio, the RDR policy.

For the eMBB users, our approach is to study the effect of each VoNR scheduling policy on the eMBB throughput region. We find that the eMBB throughput region shrinks substantially with the schedule on arrival (SOA) policy, and the various policies proposed above are able to recover almost the similar portion of the rate region (see Fig. 2), demonstrating the substantial gains of state-dependent scheduling. Next, to reduce the downlink signalling overhead and UE power consumption, we consider shorter deadlines for scheduling. We find that much of the reduction in resources used for VoNR can be achieved by reducing the deadline from 40 slots to 10 slots, with even a 2-slot deadline providing substantial gains. Preliminary simulations show that, when the channel is correlated, depending on the coherence periods, the above gains reduce.

II. RELATED LITERATURE

Resource allocation and scheduling in wireless networks have been extensively studied, particularly in the context of Long-Term Evolution (LTE) and its successor technologies. The authors of [3] focus on optimising scheduling for Voice over LTE (VoLTE) users, a guaranteed bit rate (GBR) service, alongside non-GBR data traffic. Their approach starts by formulating a joint optimal packing problem of voice packets and non-GBR data over the scheduling period equal to the voice deadline of 20 ms. This leads to a complex integer programming problem which they simplify by an optimal slot-by-slot joint scheduling of voice packets, and of non-GBR data by using the usual proportional fairness (PF) index. Finally, even this integer optimisation problem gives way to a heuristic of first packing the Physical Resource Blocks (PRBs) in a slot with voice packets, and then PF-index-based allocation of the rest of the resources to non-GBR data.

In our work, instead of a joint scheduling framework for GBR and non-GBR users, we investigate scheduling policies for Voice over New Radio (VoNR) in 5G networks, assuming that scheduling is handled independently for each user. We study optimal scheduling across the entire voice packet deadline, and formulating the scheduling problem as an optimal stopping time problem. We explore different policies, heuristic and optimal, analyzing their impact on the resources used by VoNR, and, therefore, the resources available to the eMBB flows. Furthermore, our study considers the tradeoffs between utilisation of radio resources and the savings in signalling load and UE energy, a perspective not addressed in [3]. The topics of joint scheduling of all admitted VoNR calls, and joint scheduling of the VoNR calls and eMBB flows is a topic of our current research.

In [4], the authors propose an innovative scheduling algorithm tailored for voice services in LTE networks, which considers both user perception (via the Mean Opinion Score, MOS) and network factors like Maximum Queue Size (MQS) and delay. Simulation results demonstrate that WE-MQS achieves superior performance in terms of delay and Packet

Loss Rate (PLR) compared to traditional schedulers like Frame Level Scheduler (FLS), Modified Largest Weighted Delay First (M-LWDF), and Exponential/Proportional Fair (EXP/PF). However, its performance in terms of cell throughput and fairness index is moderate, often ranking in the middle among the tested schedulers. This differs from our work from the point of view of the traffic model. We assume that VoNR packets cannot be enqueued (except for the duration of the packet deadline). Thus, we consider scheduling in which VoNR packets are handled to assure deadlines, while optimising their resource requirements, and eMBB packets are handled over the residual resources (after serving the VoNR packets) to optimise the traditional fairness utility.

In [6], a framework for efficiently managing the coexistence of ultra-reliable low-latency communication (URLLC) and eMBB traffic in 5G systems is proposed. It introduces a superposition and puncturing mechanism where eMBB traffic is allocated at the slot level, and URLLC traffic is dynamically scheduled in minislots, overlapping the eMBB allocations as needed. The study analyzes the rate losses for eMBB under three models of rate loss due to puncturing: linear, convex, and threshold, providing algorithms for joint scheduling under these loss scenarios. The authors show that optimal joint scheduling can be achieved through a gradient-based approach for eMBB and a simple placement strategy for URLLC, balancing utility maximization and latency requirements.

The authors of [6] consider the problem where URLLC traffic has such a sharp deadline that it must be accommodated within the slot in which it arrives, hence there is no scope for optimising which slot it should be transmitted in to minimise its resource utilisation. In our paper, since the voice packets do have a multislot deadline (similar to SOA, except puncturing is not considered in our work), we have examined the optimisation of VoNR resource utilisation and its impact on eMBB traffic, which uses the residual resources by the usual gradient scheduling.

III. SYSTEM MODEL

We consider a cellular network with $N + 2$ UEs connected to a gNB, out of which 2 UEs have downlink full-buffer elastic flows that require only best-effort service (called eMBB flows, henceforth), and the remaining N UEs are engaged in VoNR voice calls. The eMBB flows are motivated by downlink TCP bulk file transfers.

A VoNR packet flow in any direction is modeled as alternating between talking and listening [1]. As a worst case assumption, we take all the VoNR UEs to be in the “listening” mode, i.e., there are only downlink VoNR packet flows. Each talking VoNR flow alternates between randomly varying active speech periods (talk-spurts) and randomly varying silence periods (such as for breathing, and pauses between sentences). During a talk-spurt, the VoNR flow has 100 byte packets every 20 ms, each packet carrying coded voice, RTP/UDP/IP headers, and RLC/MAC overhead bits; in this work we have not accounted for RTP/UDP/IP header compression. The talk

spurts are modeled as geometrically distributed with a mean burst length of 400 ms, therefore, 20 voice packets; i.e., $P(\text{Burst} = k) = p^{k-1}(1-p)$, $k \geq 1$, with mean $\frac{1}{1-p} = 20$ or $p = 0.95$. The burst is then followed by a geometrically distributed silence period with a mean duration of 600 ms, or the equivalent of the time taken for 30 voice packets to arrive. In our simulations, we have considered the first packet of each VoNR call to arrive uniformly distributed over the first 40 slots, i.e., $U[1, 40]$; this stationarises the packet arrival process of each voice call.

We have, in the current work, assumed the time-frequency resources as continuous, i.e., resource allocation can be done at the level of resource elements (REs), rather than resource blocks (RBs) [5]. This makes our analytical treatment tractable, while providing useful insights. In future work, we will consider the discreteness of resources.

A. Notation

Some frequently used notation is as follows:

- Δ : The VoNR packet deadline in slots (e.g., 40 slots)
- $\Delta_i(k) := \Delta - k$ at Slot k after arrival, i.e., the remaining deadline
- $r_i(k)$: The channel rate available to a VoNR packet of UE $_i$ in slot k after its arrival, $1 \leq k \leq \Delta$

B. Model for available rates in slots

Assuming Rayleigh fading, and inverting Shannon's formula, we assume that the rate available to the i^{th} UE, denoted by UE $_i$, in successive slots is a stationary random process with the following marginal cumulative distribution function.

$$P(R_i \leq r) = 1 - e^{-\lambda_i N_0 W (2^{r/W} - 1)} \quad (1)$$

The corresponding probability density function is given as

$$f_{R_i}(r) = \lambda_i N_0 (\log_e 2) 2^{r/W} e^{-\lambda_i N_0 W (2^{r/W} - 1)} \quad (2)$$

where $\frac{1}{\lambda_i} = \bar{P}_{rx,i}$, the unfaded average downlink power received by UE $_i$.

With the above marginal distribution, we will perform simulations for independent and identically distributed (i.i.d.) rates from slot to slot, and for a simple Markov model.

IV. SCHEDULING FLOWS WITH PACKET DEADLINES

We propose and analyze four scheduling algorithms for deadline oriented packet flows.

A. The Schedule On Arrival (SOA) policy

The SOA policy operates by scheduling every VoNR packet in the first slot after its arrival, irrespective of the downlink channel conditions in that slot for the UE. This approach is efficient in terms of downlink control signalling, and UE power savings, by allowing the listener's radio to enter a sleep state for the remaining $\Delta - 1$ slots. By ignoring channel conditions, the algorithm often schedules transmissions during suboptimal slots, potentially degrading overall network performance. This

inefficiency can significantly impact the throughputs of non-real-time users, as their throughput may be reduced due to the inefficient allocation of resources.

B. Genie-Aided Scheduling (GAS)

GAS assumes perfect knowledge of the channel conditions for all Δ slots right after the arrival of a packet. With this *prescient* knowledge, the packet is scheduled in the slot in which the channel condition is most favorable, i.e., schedule UE $_i$ in the j^{th} slot such that,

$$j = \arg \max_{k \in \{1, 2, \dots, \Delta\}} r_i(k) \quad (3)$$

where $r_i(k)$ is the maximum rate at which the gNB can transmit to the i^{th} UE in the k^{th} slot after the arrival of the packet. In our simulations, this is obtained as

$$r_i(k) = W \log_2 \left(1 + \frac{\text{RSS}_i(k)}{N_0 W} \right) \quad (4)$$

where $\text{RSS}_i(k)$ is the received signal strength of the i^{th} UE in the k^{th} slot and $N_0 W$ is the noise power calculated at 300K. This approach ensures the packet is transmitted at the highest achievable rate, making it the best possible policy in terms of minimising the resources used by each VoNR flow. However, the policy is practically infeasible, and has been introduced to assess the best that any state-dependent policy can do.

C. Scheduling according to a Rate To Deadline Ratio (RDR)

The RDR policy is a heuristic approach that balances the urgency of a packet's time-to-deadline with the channel conditions in a slot. For each UE, in the k^{th} slot after the arrival of a packet, an RDR index is calculated, i.e.,

$$\gamma_i(k) = \frac{r_i(k)}{\Delta - k}. \quad (5)$$

The scheduling policy for VoNR packets of the i^{th} UE in the k^{th} slot after arrival of a packet is

$$\pi_i(k) = \mathbb{1}_{\{\gamma_i(k) > \gamma_{\text{th}}\}} \quad (6)$$

where $\gamma_{\text{th}} \in \mathbb{R}^+$ is a tunable threshold. We schedule at the smallest k , $1 \leq k \leq \Delta$, where $\pi_i(k) = 1$. Clearly, $\pi_i(\Delta) = 1$. Although this policy is simple, it is not optimal. Once we find a suitable threshold, the runtime complexity comes down to a simple comparison in each slot.

D. Optimal Stopping Time (OST) policy

The OST policy is inspired by optimal "Material Purchase (Asset Selling) with a Deadline" a standard example in stochastic optimal control, where the objective is to sequentially decide the best time to take an action to maximize returns or minimize costs. In this context, the policy evaluates the reward of scheduling a packet in each slot, with the reward dynamically decreasing as the slots progress towards the deadline. By using a dynamic programming approach, the policy recursively computes the expected cost-to-go for each slot and schedules

the packet at the slot where the trade-off between the current cost and the future opportunity cost is optimised.

If a packet reaches the last slot of its deadline without being transmitted, then it must be transmitted in that slot. Hence, the cost of scheduling the packet of UE_i in the final slot of the deadline, i.e., the Δ^{th} slot, is $J_{\Delta}^i = R_i(\Delta)$. Now, the cost in the previous slots, $k, 1 \leq k < \Delta$ can be expressed as

$$\begin{aligned} J_{\Delta-1}^i &= \max(R_{\Delta-1}^i, \mathbb{E}(J_{\Delta}^i)) = \max(R_i, \alpha_{\Delta-1}^i) \\ J_{\Delta-2}^i &= \max(R_{\Delta-2}^i, \mathbb{E}(J_{\Delta-1}^i)) = \max(R_i, \alpha_{\Delta-2}^i) \\ &\vdots \\ J_k^i &= \max(R_k^i, \mathbb{E}(J_{k+1}^i)) = \max(R_i, \alpha_k^i) \end{aligned}$$

The OST scheduling policy for UE_i in slot k is given as

$$\pi_i(k) = \mathbb{1}_{\{r_i(k) > \alpha_k^i\}} \quad (7)$$

where α_k^i is the threshold value for UE_i in slot k .

For calculating the α_k^i 's, we begin by defining $\alpha_{\Delta}^i = 0$, i.e., we have to schedule the UE in the final slot of the deadline irrespective of the rate we get. The remaining α_k^i 's are calculated as

$$\begin{aligned} \alpha_{\Delta-1}^i &= \mathbb{E}(J_{\Delta}^i) = \mathbb{E}(R_i) \\ \alpha_{\Delta-2}^i &= \mathbb{E}(J_{\Delta-1}^i) = \mathbb{E}(\max(R_i, \alpha_{\Delta-1}^i)) \\ &= \mathbb{E}\left[R_i \mathbb{1}_{\{R_i > \alpha_{\Delta-1}^i\}} + \alpha_{\Delta-1}^i \mathbb{1}_{\{R_i \leq \alpha_{\Delta-1}^i\}}\right] \\ &= \int_{\alpha_{\Delta-1}^i}^{\infty} r f_{R_i}(r) dr + \alpha_{\Delta-1}^i P(R_i \leq \alpha_{\Delta-1}^i) \end{aligned}$$

Similarly, all the α_k^i 's can be calculated recursively as follows

$$\alpha_k^i = \begin{cases} \int_{\alpha_{k+1}^i}^{\infty} r f_{R_i}(r) dr + \alpha_{k+1}^i P(R_i \leq \alpha_{k+1}^i), & \text{if } k \neq \Delta, \\ 0, & \text{if } k = \Delta \end{cases} \quad (8)$$

The OST policy represents the best achievable strategy without prior knowledge of future channel conditions, but with knowledge of the channel statistics, outperforming the heuristic policies discussed earlier. From (8), $\alpha_{k+1}^i < \alpha_k^i$, reflecting a compromise to accept poorer channel conditions as urgency grows closer to the deadline. Although this policy requires offline computations for calculating the α_k^i 's, the runtime complexity is similar to RDR.

E. Restricted deadlines

A key disadvantage of the RDR and the OST policies is that they require the UE radio to remain active for all Δ slots until the packet is scheduled, leading to higher power dissipation. To address this, a tradeoff can be made by restricting the deadline Δ , with $\Delta = 1$ corresponding to the SOA policy. This assures the UE that it can look for the next VoNR packet for up to Δ slots and then go to sleep up to when the next VoNR packet arrival is expected. This tradeoff between power efficiency and scheduling performance is explored in the next section.

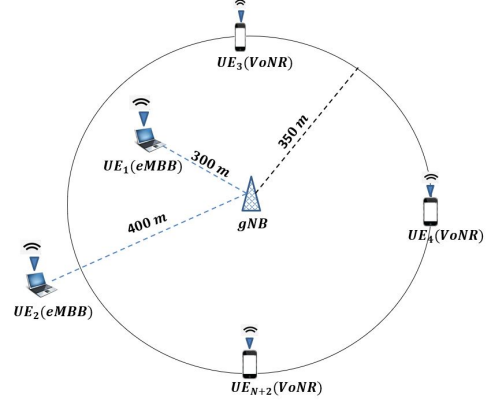


Fig. 1. Network layout, showing N VoNR UEs and 2 eMBB UEs.

V. SIMULATION RESULTS

We have used the following parameters in our simulation. The gNB transmit power is configured as 250 mW, suitable for small cell applications. The simulation timeline is divided into slots of 0.5 ms, with a deadline of 20 ms (40 slots) for each packet. The size of each VoNR packet is considered to be 100 Bytes. Two eMBB UEs are positioned at distances of 300 m and 400 m from the gNB, while the VoNR UEs are uniformly distributed on a circle of 350 m radius around the gNB as shown in Fig. 1. In any slot where a voice packet is scheduled, we assume that the remaining time-frequency resources in that slot are used for transmitting eMBB data. The system operates over a 100 MHz channel bandwidth, with a noise power of -93.8 dBm. The simulation considers 100 active VoNR calls at any given time.

The simulation results presented in Sec. V-B and Sec. V-C are for the i.i.d. available rate model, whereas the results presented in V-E are for a correlated Markov channel.

A. Performance metrics

If we only had the two eMBB UEs with full-buffer, they would have a joint rate region in $\mathbb{R}^+$, which is the set of all throughput pairs that can be sustained to those two UEs. The rate region is known to be a convex set, and also a coordinate convex set. We will display a rate region without any real-time flows, and will call it the “boundary of the rate region”. This will be the outermost rate region in the plots.

After real-time flows are included, these flows will utilise some of the radio resources, leaving only the residual resources for the eMBB UEs, thus yielding a smaller rate region. The amount of resources the real-time flows utilise will depend on the scheduling policy. Along with the “boundary of the rate region”, we will also display the rate region achieved by the various schedulers (GAS, SOA, RDR, OSTP) in the presence of real-time flows; see Fig. 2

We will also show the percentage of radio resources used by the real-time flows for the different schedulers.

Since we are not jointly scheduling of the real-time flows, it is possible that there are not enough resources to schedule all

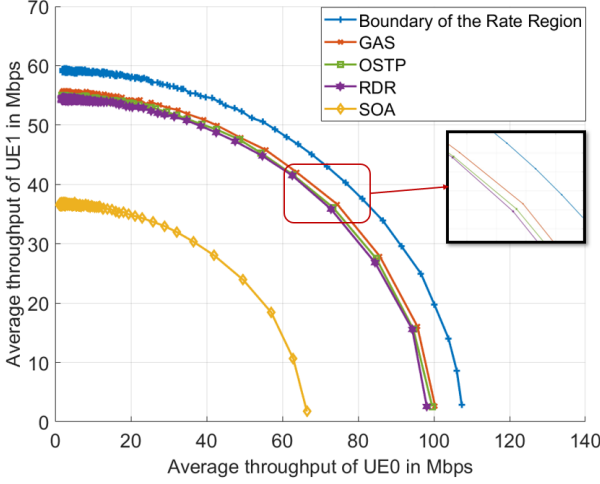


Fig. 2. Rate regions for different policies with 100 calls in the system

the packets from the real-time flows in a given slot.. We will quantify this real-time packet loss rate for each scheduler. In our ongoing research we are working on joint scheduling of all flows.

B. eMBB rate region shrinkage

Fig. 2 shows the rate regions for different scheduling policies with 100 VoNR calls. The SOA policy neglects channel conditions, leading to inefficient use of the network resources. GAS, as expected, serves as the bound, but is not practical. The RDR heuristic also performs well, after tuning its threshold, demonstrating that a simple rule-based approach can yield substantial gains. OST is implementable, and achieves the best performance without requiring future channel knowledge.

The comparison can also be seen in Table I, where the percentage of slots used up by VoNR calls decreases significantly by using RDR and OST. The improvement in the rate region

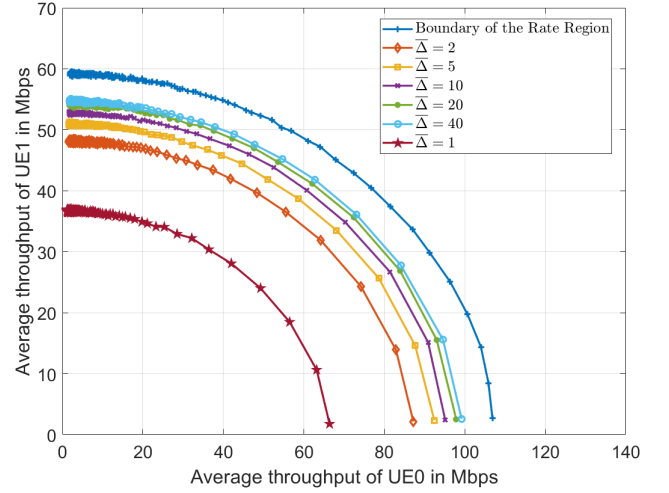
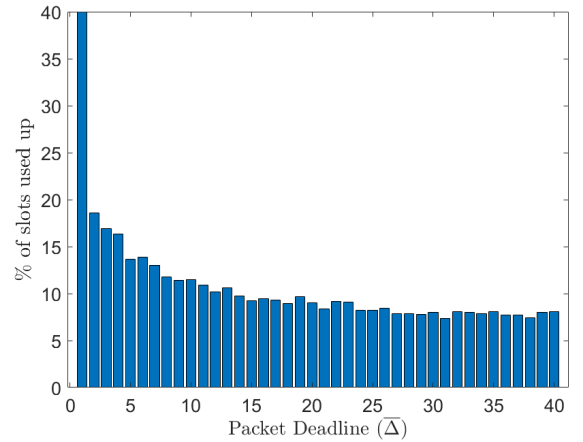
TABLE I
PERCENTAGE OF SLOT TIME USED UP BY VoNR CALLS FOR DIFFERENT NUMBER OF CALLS IN THE SYSTEM

Policy	10 Calls	50 Calls	150 Calls
SOA	20.23	38.4	52.9
RDR ($\gamma_{th} = 0.005$)	5.46	13	21.8
OST	3.8	7.9	13.5
GAS	3.22	6.53	9.75

for the eMBB UEs is significant: for UE1, the throughput increases from 38 Mbps to 55 Mbps, while for UE0, it rises from 68 Mbps to nearly 100 Mbps (In Fig. 2 compare the SOA rate region with any of the others). However, this improvement comes at the expense of keeping the UE radios active for all 40 slots, leading to increased power consumption and potentially faster battery depletion. In the next section, we will explore trade-offs between power consumption and throughput.

C. Restricted real-time packet deadlines

To permit the listening UEs to go to sleep, we explore a sub-optimal approach of OST by reducing the deadline of the


 Fig. 3. OST rate regions; varying $\bar{\Delta}$; 100 calls; i.i.d. rates over slots.

 Fig. 4. % slot time used by 100 VoNR calls; varying $\bar{\Delta}$; i.i.d. rates over slots.

packets to $\bar{\Delta}$ slots, $\bar{\Delta} < \Delta$, Δ being the original deadline. Then, the UE radio can enter sleep mode for $\Delta - \bar{\Delta}$ slots, resulting in considerable power savings. (This approach can also be applied to RDR, the results shown here are for OST).

By reducing the deadline the throughput of eMBB traffic may suffer because we forgo the opportunity to wait for potential channel improvements a better channel in later slots. We use our simulation to study the effect of varying $\bar{\Delta}$.

Fig. 3 shows the rate regions of the eMBB UEs when the VoNR UEs are scheduled with the OST policy, and i.i.d. available rates. Note that, $\bar{\Delta} = 1$ is just SOA. Just by increasing the deadline to $\bar{\Delta} = 2$, we see a significant increase in the rate region. The change in the total percentage of slots used up by the VoNR UEs is shown in Fig. 4. Finding an optimal $\bar{\Delta}$, which meets both eMBB throughput and VoNR UE power savings is an interesting problem for future work.

D. Probability of packet drops

Packet drops occur when a large number of UEs are scheduled within the same slot, causing the slot to be exhausted.

TABLE II
PACKET DROP PROBABILITY; VARIOUS POLICIES; I.I.D. CHANNEL.

Policy	100	200	250	500	1000
SOA	7	13.2	16.5	36.7	78.92
RDR	0	0	0	0.03	0.13
OSTP	0	0	0	0	0.05
GAS	0	0	0	0	0

In such cases, packets are dropped uniformly across all UEs scheduled for that slot to maintain fairness. Table II provides the simulation results for i.i.d. rates over slots. SoA is most affected by slot exhaustion since it schedules all UE packets as soon as they arrive. This issue is exacerbated in slots with poor channel capacity, further degrading performance. The packet drop probabilities for SOA would be unacceptably large, in practice. GAS achieves optimal performance by utilizing future channel information, ensuring no packet drops regardless of the number of calls in the system. RDR also performs well, with only a very small number of packets being dropped even under heavy traffic conditions with a large number of calls. OSTP outperforms RDR, minimizing packet drops further while maintaining practical feasibility.

E. Correlated available rates over slots

The analysis up to here was for i.i.d. available rates over slots. We consider a simple Markov model with the same marginal rate distribution, but with geometrically distributed coherence periods, with mean values of 5 ms (10 slots) and 10 ms (20 slots). These cases correspond to UE mobility speeds of 60 km/h and 30 km/h, which are typical driving speeds within urban environments. This approach allows us to capture the effects of mobility and channel correlation on the achievable rates, making the analysis more applicable to real-world scenarios.

From Fig. 5, we see that increasing duration of channel coherence reduces the opportunities to exploit channel variability to get a favourable channel. Still, there is a substantial gap between SOA and GAS that can be exploited by appropriate sequential decision schedulers. In future work, we will rework the OSTP analysis for this Markov channel.

VI. CONCLUSION

We studied several scheduling policies for deadline-oriented real-time traffic (such as VoNR) when sharing the downlink 5G-NR resources with elastic eMBB flows. Restricting to flow-by-flow scheduling of packet deadline calls, we have studied four policies: SOA, GAS, RDR, and OSTP. SOA is similar to the policy that is used in practice, commits too soon, and ends up using a lot of radio resources. The other policies, of which RDR is the most practical, perform quite similarly in terms of the eMBB rate region achieved (see Fig. 2), but need additional downlink signalling, and do not permit the UEs to go to sleep. By restricting the deadlines, we can achieve tradeoffs between resource utilisation by the real-time calls, and the time for which the real-time UEs can sleep; see Fig. 3 and Fig. 4.

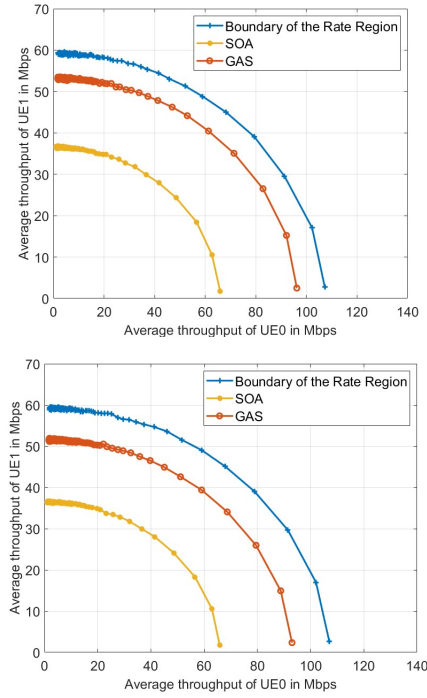


Fig. 5. eMBB rate regions for coherence times 5 ms (top) and 10 ms (bottom).

Since we schedule the real-time packets independently across the calls, there is a possibility of packet loss, which can be substantial for SOA, limiting its VoNR carrying capacity; see Table II. We also provide a brief study of the case of correlated rate over slots. In future work, we will explore joint scheduling of all VoNR UEs as well as joint scheduling of both VoNR and eMBB UEs

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REFERENCES

- [1] J. Daigle and J. Langford, "Models for analysis of packet voice communications systems," in *IEEE Journal on Selected Areas in Communications*, vol. 4, no. 6, pp. 847-855, September 1986.
- [2] Bertsekas, Dimitri P. *Dynamic Programming and Optimal Control*. United States: Athena Scientific, 2012.
- [3] M. Mohseni, S. A. Banani, A. W. Eckford and R. S. Adve, "Scheduling for VoLTE: Resource Allocation Optimization and Low-Complexity Algorithms," in *IEEE Transactions on Wireless Communications*, vol. 18, no. 3, pp. 1534-1547, March 2019.
- [4] Duy-Huy Nguyen, Hang Nguyen, Éric Renault. WE-MQS: "A new LTE Downlink Scheduling Scheme for Voice Services based on User Perception". *International Journal of Computer Applications*, 2016, 142 (10), pp.28-36
- [5] K. Takeda, H. Xu, T. Kim, K. Schober and X. Lin, "Understanding the Heart of the 5G Air Interface: An Overview of Physical Downlink Control Channel for 5G New Radio," in *IEEE Communications Standards Magazine*, vol. 4, no. 3, pp. 22-29, September 2020
- [6] A. Anand, G. De Veciana and S. Shakkottai, "Joint Scheduling of URLLC and eMBB Traffic in 5G Wireless Networks," *IEEE INFOCOM 2018 - IEEE Conference on Computer Communications*, Honolulu, HI, USA, 2018, pp. 1970-1978.